AI Usage Log

YZM5109 Large Language Models – Assignment I (Option C)
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1. Tools used

Tool	Use
Claude (Anthropic), through Claude Code in a cloud session	Experiment design, all Python code, post-hoc analysis, report text, this log, the submission package.
Google Colab (A100 GPU)	Running the experiments. Run by the student; not an AI tool.

2. Key prompts and outputs

The student wrote the prompts in Turkish; they are translated here.

1. **Prompt:** "Can you see and explain this assignment PDF? How should we do it? Could you handle the work overnight via Colab and Hugging Face?" The PDF link could not be opened from the AI session, so the student pasted the assignment text.
   Output: Choice of Option C with Qwen2.5-1.5B-Instruct and GSM8K, and four conditions: direct, zero-shot chain-of-thought, PAL, and LoRA evaluated with the direct prompt. The AI also produced a self-contained Colab notebook, a helper library, a CPU smoke test with a tiny random model, and an IEEE report draft. The AI session had no GPU and no access to Hugging Face or Colab, so it could not run the real experiment.
2. **Prompt:** The student uploaded the results of a 20-item pipeline check run on Colab.
   Output: The AI found a bug in its own PAL scorer. Programs ending in a bare expression, such as `total_cost`, were scored as "no output". The scorer was fixed to print the final bare expression, and regression tests were added. *Disclosure:* this check used 20 test-split items. Only the answer-parsing code changed; prompts, decoding settings and hyperparameters were not changed.
3. **Prompt:** "Make sure the notebook is error-free." and "Colab Pro lets us use L4 or A100."
   Output: The notebook was hardened with a GPU check, automatic batch size, resume-safe rescoring and real-run defaults, and A100 was recommended.
4. **Prompt:** The student uploaded the full-run results zip.
   Output: Post-hoc analysis script covering arithmetic error rate, accuracy by problem length, complementarity and a PAL sensitivity check. The AI also completed the report and compiled it to PDF.
5. **Prompt:** The student uploaded this assignment form: "Prepare the file I will submit according to it."
   Output: The AI checked the submission against the form. It added a second metric (answer validity rate), prediction CSVs, an evaluation script with a terminal screenshot of the scores, configuration and prompt files, error examples, a prompt appendix, this log and the ZIP package.

3. What the student did

- Chose to follow the proposed design and selected the GPU type.
- Ran the pipeline check and the full experiment on Google Colab (NVIDIA A100-40GB) and supplied all result files.

- Reviewed the code, the results and the report before submission.

4. Verification

- All numbers in the report come from the saved per-item predictions of the Colab run. They are recomputed by `evaluate.py` and `analysis/deep_analysis.py`, and `screenshots/` shows that output.
- The pipeline was tested end to end on CPU with a tiny random model, including the resume path, before it ran on GPU.
- Concrete model failures quoted in the report were read from the raw outputs, not generated.